

HR ANALYTICS & ATTRITION INTELLIGENCE

Complete Interview Preparation Guide & Project Walkthrough (End-to-End)

1. Project Executive Summary (Samajhne Ke Liye Overview)

Is Project Ka Main Maqsad Kya Hai?

Yeh project ek enterprise-level HR Analytics system hai. Companies mein employees company kyun chhodte hain (ise **Attrition** ya **Employee Turnover** kehte hain), iska data analyze karke HR Management ko decision-making mein help karna is project ka main objective hai.

Key Insights Discovered from Data (Main Findings):

- **Total Employees Analyzed:** 1,470 records (IBM HR Dataset)
- **Overall Attrition Rate:** 16.12% (237 employees left the company)
- **Biggest Attrition Driver:** Overtime Work! Overtime karne wale employees ka attrition **30.53%** hai, jabki non-overtime employees ka sirf **10.44%** hai (Nearly 3X difference!).
- **Highest Attrition Department:** Sales Department (20.63% attrition rate), followed by HR (19.05%) and R&D; (13.84%).
- **Machine Learning Model Accuracy:** Random Forest Classifier achieved **0.8127 ROC-AUC Score** (81.3% accuracy).

2. Complete Tech Stack Used & Why (Kaunsa Tool Kyun Use Hua)

Technology / Tool	What it Does in This Project	Key Concepts Used
SQL (T-SQL)	Data cleaning, deduplication, creating views, and advanced window functions.	CTEs, ROW_NUMBER(), NTILE(4), CREATE VIEW, Aggregations
Python & Pandas	Automated Data Pipeline, EDA, statistical checks, data formatting.	read_csv(), groupby(), apply(), Risk Scoring
Scikit-Learn (ML)	Predicting employee turnover risk using machine learning.	RandomForestClassifier, ROC-AUC (0.8127), Feature Importance
Streamlit & Plotly	Building an interactive web dashboard application (app.py).	px.bar(), px.scatter(), sidebar filters, live metrics
Power BI & DAX	Business Intelligence dashboard & dynamic filter context reporting.	DAX Measures, CALCULATE, DIVIDE, KPI Cards, Slicers

3. Detailed Walkthrough of Project Files (Har File Ka Kaam)

A. SQL Scripts Suite (sql/ folder)

- sql/01_data_cleaning.sql:** Clean karta hai dataset ko. Key technique: Duplicate rows hatane ke liye CTE aur ROW_NUMBER() OVER (PARTITION BY EmpID ORDER BY EmpID) use kiya gaya hai. Foreign characters/typos standardise kiye gaye hain.
- sql/02_analytical_views.sql:** Power BI aur Reporting ke liye 6 modular views banaye gaye hain (e.g. vw_Attrition_By_Age, vw_Attrition_By_Department, vw_Attrition_By_Overtime).
- sql/03_advanced_queries.sql:** Advanced queries implement ki hain. Example: Income Quartiles divide karne ke liye NTILE(4) OVER (ORDER BY MonthlyIncome ASC) use kiya gaya hai.

B. Python Scripts & Machine Learning (scripts/ folder & app.py)

- scripts/hr_analytics_pipeline.py:** Automated ETL script. Data quality check karta hai aur retention risk scoring compute karta hai.
- scripts/train_attrition_model.py:** Machine Learning model script. Random Forest Classifier train karta hai (81.3% ROC-AUC score). Output mein Feature Importance calculate karke top drivers (Age, Income, Overtime) rank karta hai.
- app.py:** Interactive Streamlit Web App application. User browser me sliders/dropdowns change karke live graph dekh sakta hai.

C. Reports & Technical Documentation (reports/ folder)

- reports/DAX_Measures_Reference.md:** Power BI DAX formulas ka documentation (Total Employees, Attrition Rate %, Overtime Attrition %).
- reports/HR_Attrition_Executive_Report.md:** Executive business insights report and HR leadership retention recommendations.
- reports/Interview_Preparation_Guide.md:** STAR method interview Q&A; guide.

4. Key Interview Questions & How to Answer (Interview Q&A;)

Q1. Tell me about this HR Analytics project on your resume.

How to Answer (60-sec Pitch):

"In this project, I built an end-to-end workforce intelligence solution analyzing 1,470 employee records to identify key drivers of attrition. I used SQL (CTEs, Window Functions, Views) for data cleaning, Python (Pandas & Scikit-Learn) for an automated ETL pipeline and a Random Forest classification model (ROC-AUC 0.8127), and Power BI / Streamlit for dynamic visualization. A major finding was that Overtime employees experienced a 30.5% attrition rate compared to 10.4% for non-overtime staff."

Q2. How did you handle duplicate data in SQL?

Technical Answer:

"I used a Common Table Expression (CTE) with ROW_NUMBER() OVER (PARTITION BY EmpID ORDER BY EmpID). Any row with RowNum > 1 was identified as a duplicate and safely deleted, ensuring 100% data integrity."

Q3. What machine learning model did you use, and how did you evaluate it?

Technical Answer:

"I trained a Random Forest Classifier using Scikit-Learn with balanced class weights to handle class imbalance (since attrition is ~16%). I evaluated the model using ROC-AUC score (achieving 0.8127) and analyzed Feature Importances, which revealed Monthly Income, Age, and Overtime as the top predictors of employee departure."

Q4. How did you handle DAX measures in Power BI?

Technical Answer:

"I created explicit DAX measures using CALCULATE(), DIVIDE(), and COUNT(). For example: Attrition Rate % = $\text{DIVIDE}([\text{Attrition Count}], [\text{Total Employees}], 0) * 100$. Explicit measures allow dynamic recalculation across all dashboard slicers."

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